Computer Programming with Python

Dr. Aamir Alaud Din

Institute of Environmental Sciences and Engineering (IESE)
School of Civil and Environmental Engineering (SCEE)
National University of Sciences and Technology (NUST), Islamabad

Topic 7 - Functions, Python Packages, Strings, and Lists

October 21, 2024

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

RECAP

- Values of elements in a list can be reassigned but not in tuples and sequence generated by `range` function.
- The `bool` data type has only two values which are `True` and `False` based on which decisions are made in computer programming.
- The `None` data type assigned to a variable has no value stored in it and arithmetic operations are invalid on this data type.
- This data type is important to deal with missing values in data files.
- Assignment operators are used to update the values stored in variables.

RECAP

- Comparison operators are used to compare to variables and they compare equality, not equality, greater than, less than, greater than or equal to, and less than or equal to cases of variables.
- The logical operators combine the results of relational operators and the result is a `bool` data type to determine the processing of a piece of code in the program.
- Truth tables are important and clarify the outcome of all logical operations.
- Membership operator helps to determine if a variable is contained or belongs to a main list or not.
- The identity operator checks if the `id` of two variables are same (`True`) or not (`False`).

RECAP

- The operators are common to all computer programming language because they are very important in taking decisions in computer programming.
- One big problem with operators is that they don't produce bug report and programs terminates normally and we feel no problem in the program.
- However, all the programs (even big companies) test the results of their programs because despite of no bugs, wrong usage of operators leads to wrong results.
- Rectifying errors in such cases is tedious and challenging and therefore you must have a sound knowledge of operators.
- It is advised to practice data types and operators a lot to write good computer programs.

TABLE OF CONTENTS

RECAP
OBJECTIVES
THE WHY SECTION
FUNCTIONS
PYTHON PACKAGES, METHODS, AND ATTRIBUTES
STRINGS
LISTS
SUMMARY

OBJECTIVES

After taking this lecture and studying, you should be able to

- Describe functions in your own words and identify functions in any python program.
- Describe in your own words the difference among functions, methods, and attributes.
- Know about python packages and then import and use python packages.
- Use string methods and attributes for string manipulation.
- Use list methods and attributes for list manipulation and use.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

THE WHY SECTION

- With the fast growing development in IT and interdisciplinary knowledge, many of us cannot write computer programs of some other fields.
- As an example, an environmental engineer can study environmental engineering topics on atomic and molecular level using molecular dynamics simulations without having a sound knowledge of molecular dynamics simulations.
- How come is it possible?
- Withoug having a complete knowledge of calculus, analytical geometry, differential equations, linear algebra, and numerical methods to a level where you can code everything about these subjects, you can use these mathematical techniques to solve many problems.
- How is it possible?

THE WHY SECTION

- How can we manipulate strings, for example, converting lowercase words to uppercase words without wrinting full fledge programs?
- Similarly, how can we convert string to title case, how to find strings with a prefix or suffix, and many other such operations on strings without writing the full programs?
- How can we manipulate lists, for example, adding entries on specified locations without writing computer programs to encapsulate the complete methodology?
- How to join two lists, append elements to a list, and many other such functions without thinking of the algorithm of implementation?
- We will answer to all these questions shortly.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

FUNCTIONS

- A python functions is a reusable computer program.
- The functions always start with a keyword (string) following paranthesis.
- The best example of function is the `print` function.
- Note that, you just provide input to the `print` function without knowing an programming to show some message on standard output display.
- You just give input (input arguments) within paranthesis and the rest of the job is of `print` function.
- A function may take no input arguments to several input arguments.
- The `unit_vector` function to find unit vector of a given vector will clarify the usage and importance of functions.

FUNCTIONS

- From `unit_vector` function, it is clear that without writing the complete program, we can find the unit vector of a given vector by just giving input vector the the function.
- So, functions help in computations without complete knowledge of computation and we just need to give input and get output.
- This is the reason, functions are called the reusable programs which can be reused by many users.
- We will learn to write our own functions in the upcoming topics.
- So far, it is sufficient to identify functions in any python programs.
- Whenever, you encounter a keyword following parenthesis, you should come to know that this is a function.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- A python package is a collection of several functions related to a single field of study.
- For example math package in python doesn't provide a single math function, but several math functions like trigonometric functions, exponential function, and logarithmic function etc.

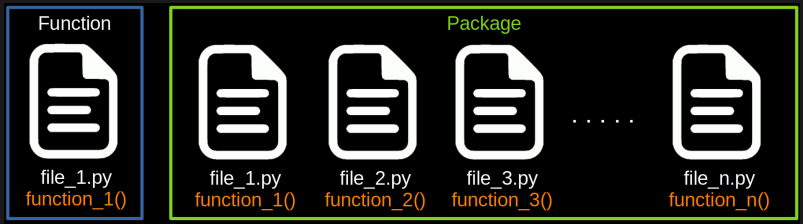

Figure. Python function vs package.

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- A single function within a package is called a method.
- In other words, a method is also a function within a package.
- As method is also a function and we know that all functions require no input to several inputs.
- A function named `test` with no input will be called as shown below.

```
>>> test()
```

- Another function named `job` with three inputs `x`, `y`, and `z` will be called as shown below.

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

```
>>> job(x, y, z)
```

- Since a method is also a function available in a package, so a method may also take no to several inputs.
- Recall the `complex` data type in numeric data type.
- Let's define a complex number as shown below.

```
>>> x = 2 + 3j
```

- We can find the conjugate of `x` by calling the `conjugate` function as shown below.

```
>>> x.conjugate()
```

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- If we assign conjugate of `x` to a variable `y` and print it, the code for this purpose will be as follows.

```
>>> y = x.conjugate()
>>> print(y)
2 - 3j
```

- Note the use of paranthesis after the keyword `conjugate`.
- What if we don't use the paranthesis after the keyword `conjugate` as shown below?

```
>>> y = x.conjugate
```

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- The above line of code will generate the conjugate object without assigning the value of `2 - 3j` to variable `y`.
- Printing of `y` will show the object generation but not the conjugate *i.e.*, `2 - 3j` as shown below.

```
>>> print(y)
<built-in method conjugate of complex object at
0x71f75129ea50>
```

- It means, paranthesis are mandatory in functions and methods even if no input is passed to the method/function.
- Now recall that we also found the real and imaginary parts of the complex number using the keywords of `real` and `imag`.

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- No paranthesis were required to find the real and imaginary parts of the complex number as shown below.

```
>>> x.real
>>> x.imag
```

- If we assign the real and imaginary parts to the variables `y` and `z` and print them, the values will be printed (no paranthesis required) as shown below.

```
>>> y = x.real
>>> z = x.imag
>>> print(y)
2.0
>>> print(z)
3.0
```

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- When a job is accomplished by a keyword without requiring paranthesis, this is not a function/method and is known as the attribute.
- So, functions/methods require paranthesis but not the attributes.
- What if paranthesis are used with attributes, as shown below?

```
>>> y = x.real()
```

- The above line of code, instead of assigning real part of complex number to variable `y` will pop up error message.
- Both functions/methods and attributes perform some job but functions/methods require paranthesis with no to several inputs while attributes don't require paranthesis.

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- Many functions which we use in python programs without calling the package are available in python's standard library which loads automatically so that we use these functions directly.
- The examples of functions/methods available in python's standard library are `print`, `range`, and `conjugate` functions.
- If we want to use the tigonometric functions like `sin` function to find the sine of a given angle, can we use the following line of code?

```
>>> y = sin(2.1)
```

- The answer is, NO.
- The trigonometric functions are not available in python's standard library.

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- In order to use these and many other such functions, we need to import the relevant package.
- Sometimes, it is also possible to load the required function/method from the package in order to use that function/method.
- As already discussed, the PSF provides its own developed packages *click here to see*.
- There are also packages which are provided by different universities and organizations.
- These packages are available on PyPI *click here to see*.
- It is time to learn how to use packages.

Math Package: Importing without Naming

- Suppose we want to use the `sin` function to find the sine of a given angle.
- The `sin` function is available in math package.
- In order to use `sin` function from math package, we have to import math package.
- The code to use `sin` function from math package is shown below.

```
>>> import math
>>> y = math.sin(2.1)
>>> print(y)
```

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- The output of the above code is shown below.

```
0.8632093666488737
```

- If we want to use the trigonometric identity $sin^2\,\theta + cos^2\,\theta = 1$, the program will look as shown below.

```
>>> import math
>>> y = (math.sin(2.1))**2 + (math.cos(2.1))**2
>>> print(y)
1.0
```

Do Yourself

Write a python program to evaluate the following formula

$$y = tan\,\theta + e^{sin\,\theta}$$

Math Package: Importing with Name

- Another way of importing a package is to call it by name of your own choice, for example, calling `math` package by the name of `m`.
- The trigonometric identity can also be verified by using the following program.

```
>>> import math as m
>>> y = (m.sin(3.2))**2 + (m.cos(3.2))**2
>>> print(y)
1.0
```

- Instead of `m`, you can call the package with the name of `aamir` and then `sin` function will be called by `aamir.sin()`.

Math Package: Importing Functions Directly

- We can also call the relevant functions directly from a package, for example, calling `sin` and `cos` functions fro `math` package to verify trigonometric identity will look like this.

```
>>> from math import sin, cos
>>> print(sin(6.75)**2 + cos(6.75)**2)
1.0
```

- Since we imported only `sin` and `cos` functions, so we can only use them and cannot use any other function like `tan` and `exp` functions.
- In order to call multiple functions from a package, separate functions by commas as shown above.

Math Package: Importing Functions by Names

- Just like calling a package by name, we can call the functions by names.
- An example of verifying trigonometric identity by calling `sin` and `cos` functions by the names `s` and `c` is given below.

```
>>> from math import sin as s, cos as c
>>> print(s(5.2)**2 + c(5.2)**2)
0.9999999999999999
```

- Again, we can use only those functions which we called from the package and not other functions which are available in the package being used.
- Calling multiple functions with names still requires separation of functions by commas as shown above.

Math Package: Importing All Functions

- We can call all the functions from `math` package and use them directly as shown in the verification of trigonometric identity below.

```
>>> from math import *
>>> print(sin(1.6)**2 + cos(1.6)**2)
1.0
```

- The method to call all function from the package is a good idea to use any function from the package.
- This method allows the use of any function available in the package without explicitly calling it.

Math Package: Recommendations

- Although all the methods shown above can be used to accomplish a give computation.
- But, there is a little problem if a function is available in different packages and we have to use all the packages like the scenario shown below.

```
>>> from numpy import *
>>> from math import *
>>> print(sin([1.0, 2.0, 3.0]))
```

- The `sin` function is available in `numpy` as well as `math` package.
- In the above code, which `sin` function is called?

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

- The `sin` function from `numpy` or `sin` function from `math` package.
- In order to avoid this confusion, it is highly recommended to import packages by names and use a `package.function` method to be aware of the function call from the known package as shown below.

```
>>> import numpy as np
>>> import math as m
>>> print(m.sin(np.array([1.0, 2.0, 3.0])))
```

- This method of call clearly specifies that `sin` function is called from `math` package and `array` function is called from `numpy` package.
- It is highly recommended to use math package to verify at least five trigonometric identities of your own choice.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

STRINGS

- We learnt indexing and slicing of strings.
- It is time to learn and use string methods to do something useful.
- Lets define a string and convert it to uppercase and to test if a string is comprised of digits.

```
>>> x = "aamir"
>>> y = "299792458"
>>> print(x.upper())
AAMIR
>>> print(y.isdigit())
True
```

- Define strings and test various string methods from page 34 of reference book 1.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

LISTS

- Let's define an empty list and use list methods to accomplish some tasks.

```
>>> l = []
>>> l.append(10)
>>> l
[10]
>>> l.extend([20, 30, 40, 50])
>>> l
[10, 20, 30, 40, 50]
>>> l.insert(5, 60)
>>> l
[10, 20, 30, 40, 50, 60]
```

- Try list methods shown on page 47 of reference 1.

TABLE OF CONTENTS

RECAP

OBJECTIVES

THE WHY SECTION

FUNCTIONS

PYTHON PACKAGES, METHODS, AND ATTRIBUTES

STRINGS

LISTS

SUMMARY

SUMMARY

- A function is a reusable python program which may require no to several inputs.
- A python package is a collection of functions which can be utilized to accomplish a given computation/task.
- Python functions/methods and attributes do something needful for the user but functions/methods require paranthesis with no to several inputs while attributes don't require paranthesis and input.
- Python packages can be imported with and without names.
- If a package is called without name, the name of package dot the function is used to call the function from the package.

SUMMARY

- If a package is imported with name, the name dot function can be used to call and use a function from the package.
- The specific functions from a given package can be called with and without names separated by commas.
- All functions from a package can be called from a package by a $\star$ symbol.
- In order to keep track of the functions and package used, it is recommened to call packages by name.
- String and list methods allow user to accomplish several useful tasks.
- It is highly recommended to import and use `random` and `statistics` packages to accomplish some tasks/computations.

THANK YOU